

Recipe Optimizer

A new autonomous mode that sits beside the existing Standard optimizer. The user defines a recipe once; the system generates templates, runs NinjaTrader, ingests results, selects winners, and feeds them into the next stage — all the way to final validation.

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03	Autonomous Run Dashboard	04	Stage Results & Buckets
05	Child Lineage / Branch Inspector	06	Final Review / Decision Dashboard

RECIPE OPTIMIZER · NEW AUTONOMOUS MODE

Set up the recipe

Define the recipe once. The system generates templates, runs NinjaTrader, ingests results, selects winners, and feeds them into the next stage — all the way to final validation.

Save draft

Continue: Matrix & Roles →

STAGE PLAN · WHAT THE SYSTEM WILL DO

Each stage feeds the next. You can pause, branch, or override between stages.

Autonomous research loop

1

Broad Bucket Search

QUEUED

- System generates matrix templates
- System runs NinjaTrader
- System ingests results
- System selects top candidates per bucket

Templates 12 Combos ≈ 9.6K

2

Child Refinement

QUEUED

- System creates child templates from selected parents
- System refines around winner values
- System runs child templates
- System selects refined candidates

Templates ≈ 24 Combos ≈ 4.8K

3

Risk / Daily Behavior

QUEUED

- System optimizes risk parameters
- System runs risk templates
- System filters by drawdown & trade cadence

Templates ≈ 6 Combos ≈ 1.2K

4

Final Backtest Validation

QUEUED

- System creates fixed final-backtest templates
- System runs validation
- System prepares deployment package

Templates ≈ 4 Combos ≈ 80

RECIPE

Inputs & goals

Strategy

Seed template (starting parameter values)

PantheonMasterBotV01TesterV2 · 30 params

Pass1_Breakout_Start00 · breakout · NQ 06-26

C:\Users\Owner\Documents\NinjaTrader 8\bin\Custom\Strategies\PantheonMasterBotV01TesterV2.cs

Recipe name

Pantheon NQ — Time × Reverse × Slow MA

Contract

NQ 06-26

Market suffix

NQ

Target final candidates (deployable strategies)

4

System keeps trying until it has at least this many that pass final validation

Max total runtime (hours)

12

System pauses if exceeded

Max combinations (safety cap)

250,000

Stops if total scope exceeds this

ESTIMATED SCOPE

All artifacts saved to session/rec_8c2a44/

Across all 4 stages

TEMPLATES

≈ 46

generated across all stages

COMBINATIONS

≈ 15.6K

total backtests planned

EST. RUNTIME

3.4 hr

at 80ms / backtest

FINAL CANDIDATES

≥ 4

goal

⚠ **Heads up.** The system will autonomously run NinjaTrader. Make sure NT8 is open and the batch AddOn is loaded before clicking **Start Recipe**. You can pause or stop between stages at any time.

RECIPE OPTIMIZER · STEP 2 OF 4

Matrix axes & parameter roles

Matrix axes split the search into separate optimizer jobs. Every other parameter gets a role telling the system whether to sweep it, fix it, or carry it forward from a parent winner.

← Back

▶ Start Recipe

A · MATRIX AXES

Each combination of axis values becomes its own optimizer job. The system runs them in parallel where possible.

Variables that create separate templates

⋮

StartTimeH

Trading-window start hour

0

4

8

12

16

20

+ add

× 6

VALUES

×

⋮

Reverse

Trade direction flip

false

true

+ add

× 2

VALUES

×

⋮

averageSlow

Slow MA family · optional

50

100

150

200

250

300

350

400

+ add

× 8

VALUES

×

+ Add matrix axis

6

×

2

×

8

=

96

START HOURS

REVERSE

SLOW MA

STAGE 1 TEMPLATES

B · PARAMETER ROLES

Filter parameters...

What the system does with each parameter

● Matrix axis

Splits into separate templates

● Optimize

Swept inside each template

● Refine

Child stage sweeps around parent winner

● Inherit

Carried forward from parent winner

● Fixed

Constant for the whole recipe

● Validation only

Used only in the final backtest

PARAMETER	TYPE	ROLE	BEHAVIOR
StartTimeH	int	● Matrix axis	6 axis values
DurationTimeH	int	● Fixed	4
Reverse	bool	● Matrix axis	false, true
averageFast	int	● Optimize	2 → 10 step 1 · 9 values
averageSlow	int	● Matrix axis	8 axis values
MaxStop	double	● Optimize	50 → 400 step 25 · 15 values
MaxTPRatio	double	● Refine	Refine ±20% around parent winner
ProfitStop	double	● Inherit	Stage 3 tunes from Stage 2 winner
LossStop	double	● Inherit	Stage 3 tunes from Stage 2 winner
MaxTrades	double	● Inherit	Stage 3 tunes from Stage 2 winner
Contracts	int	● Validation only	Final-stage backtest at size 1
Long	bool	● Fixed	true
Short	bool	● Fixed	true

How the system picks winners between stages

Applied after each stage. Surviving candidates become parents for the next stage.

GROUPING & KEEP

Group by

matrix axes

StartTimeH, Reverse, averageSlow

Keep per group

Top 2

Ranking method

DD-adjusted score

Tie breakers

lower DD → higher trades → higher net

FLOORS / CEILINGS

Min trades

≥ 40

Min profit factor

≥ 1.50

Max drawdown

≤ \$2,500

Min net profit

≥ \$3,000

RECIPE OPTIMIZER · RUNNING

Autonomous run

The system is working through stages without your input. Pause or stop between templates at any time.

Pause after current template

Stop

CURRENT STAGE

2 of 4

Child Refinement

TEMPLATES COMPLETED

103

of ≈ 124 total

ROWS PARSED

8,640

98.6% successful

RUNTIME

51m

ETA ≈ 2h 14m remaining

✓

Broad Bucket Search

DONE

✓

System generates matrix templates

✓

System runs NinjaTrader

✓

System ingests results

✓

System selects top candidates per bucket

Templates 96/96 Selected 18

2

Child Refinement

RUNNING

✓

System creates child templates from selected parents

✓

System refines around winner values

○

System runs child templates

○ System selects refined candidates

Templates 7/18 ETA 42m

3

Risk / Daily Behavior

QUEUED

○

System optimizes risk parameters

○

System runs risk templates

○

System filters by drawdown & trade cadence

Templates ≈ 6 Combos ≈ 1.2K

4

Final Backtest Validation

QUEUED

○

System creates fixed final-backtest templates

○

System runs validation

○

System prepares deployment package

Templates ≈ 4 Combos ≈ 80

NOW RUNNING

Stage 2 · Child Refinement

○

System is running NinjaTrader · template P2_T007_S08_R0_M200_refine · chunk 2 of 4

NT8 status:

heartbeat 2s ago

 State: Running NinjaTrader Last update: 11:42:31 Failed templates: 0

View Stage 1 results

Rerun current template

Skip this template

ACTIVITY

What the system has done

View full log

11:42:08

◆

Stage 2 · Child Refinement started. 7 child templates generated from Stage 1 winners.

11:41:55

✓

Stage 1 complete. 96 templates · 8,640 rows scanned · 18 winners selected across 12 buckets.

11:41:42

!

2 boundary winners in buckets Start=20 · Reverse=true — Stage 2 will expand the range.
Surface so you can decide whether to widen the matrix axis instead.

11:38:11

◆

Bucket Start=08 · Reverse=false · Slow=200 → top row PF 2.31 selected as parent.

11:24:50

▶

NinjaTrader running template P1_T072_S20_R1_M350 · chunk 3 of 5.

11:09:32

✓

Template P1_T058_S12_R0_M250 finished · 540 rows · best PF 1.92.

10:51:04

▶

Recipe started. Stage 1: 96 matrix templates generated.

! 2 boundary winners in Stage 1 — winning values landed at the edge of their range. Stage 2 will automatically widen those axes by 25%, but you can also widen them yourself.

Inspect boundary buckets

Widen axes manually

OVERRIDE CONTROLS

↺ Rerun current stage from scratch

↵ Pause after this stage

⊕ Promote a row manually as parent

⊖ Reject a selected candidate

OUTPUT LOCATION

sessions / rec_8c2a44 /
└ stage_1 / templates · results · selected.csv
└ stage_2 / templates · (running)
└ recipe.json
└─ run.log

STAGE 1 · BROAD BUCKET SEARCH · COMPLETE

Stage results & buckets

The system grouped 8,640 result rows into 12 buckets (one per matrix combination) and picked the top candidate per bucket using your selection rules. These become parents for Stage 2.

← Back to run

Export bucket CSV

Rerun stage

TEMPLATES

96

all completed

OPTIMIZER ROWS

8,640

98.6% parsed cleanly

PASSING GUARDRAILS

412

of 8,640 rows

SELECTED

18

2 per bucket × 12 buckets, 6 short

STAGE BEST-OF

Across all 12 buckets

BEST PF

2.31

Start=08 · Rev=false · Slow=200

BEST NET

\$18,420

Start=08 · Rev=false · Slow=200

BEST SCORE

9,821

DD-adjusted, same bucket

WORST DD

-\$2,480

Start=20 · Rev=false · Slow=350

! 2 boundary winners. Buckets Start=20 · Reverse=false · Slow=350 and Start=20 · Reverse=true · Slow=400 picked rows at the edge of the swept ranges. The system will widen them by 25% for Stage 2 — you can override below.

BUCKETS

One row per matrix combination · 12 buckets

All 12

Passing 10

No candidate 1

Boundary 2

Q Filter buckets...

BUCKET	BEST PF	BEST NET	DD	TRADES	SELECTED CANDIDATE	CHILD STAGE	STATUS	
Start=00 · Reverse=false · Slow=150 720 rows · 720 parsed	1.42	\$6,240	-\$1,810	162	P1_T003 row #4012	↓ P2_T001	Selected	...
Start=00 · Reverse=true · Slow=200 720 rows · 720 parsed	1.18	\$2,810	-\$2,440	198	—	—	No Candidate	...
Start=04 · Reverse=false · Slow=200 720 rows · 720 parsed	1.84	\$9,120	-\$1,620	188	P1_T009 row #7724	↓ P2_T002	Selected	...
Start=08 · Reverse=false · Slow=200 720 rows · 720 parsed	2.31	\$18,420	-\$1,840	412	P1_T021 row #1024	↓ P2_T003	Selected	...
Start=08 · Reverse=false · Slow=250 720 rows · 720 parsed	2.18	\$17,110	-\$1,620	388	P1_T022 row #2210	↓ P2_T004	Selected	...
Start=08 · Reverse=true · Slow=200 720 rows · 720 parsed	1.62	\$8,410	-\$2,010	244	P1_T025 row #5102	↓ P2_T005	Selected	...
Start=12 · Reverse=false · Slow=175 720 rows · 720 parsed	2.05	\$16,240	-\$1,980	401	P1_T033 row #9821	↓ P2_T006	Selected	...
Start=12 · Reverse=true · Slow=200 720 rows · 720 parsed	1.71	\$9,810	-\$2,200	280	P1_T036 row #4422	↓ P2_T007	Selected	...
Start=16 · Reverse=false · Slow=300 720 rows · 720 parsed	1.86	\$14,210	-\$2,240	367	P1_T045 row #6611	↓ P2_T008	Selected	...
Start=20 · Reverse=false · Slow=350 720 rows · 720 parsed	1.94	\$11,280	-\$2,480	312	P1_T067 row #3018	↓ P2_T009	Boundary	...
Start=20 · Reverse=true · Slow=400 720 rows · 720 parsed	1.76	\$9,820	-\$2,380	296	P1_T072 row #2240	↓ P2_T010	Boundary	...

OVERRIDE

Per-bucket actions

Rerun bucket with widened axis

Choose

Manually promote candidate override selection

Choose

Reject candidate drop from Stage 2

Choose

CHILD LINEAGE · BRANCH INSPECTOR

How candidate #C-04 was discovered

Every final candidate traces back to a Stage 1 bucket. Use this view to understand why a candidate exists and to branch, clone, or reject any node in its lineage.

← Back to review

#C-01 Start=12 · Rev=false · Slow=175

Compare candidates

LINEAGE

Click any node to inspect it on the right. Right-click for rerun / clone / reject.

Recipe → bucket → parent → child → final

●

RECIPE rec_8c2a44 · Pantheon NQ — Time × Reverse × Slow MA
3 matrix axes · 13 parameters · 4 stages

●

STAGE 1 · BUCKET b_S08_R0_M200
Selected
StartTimeH=8 · Reverse=false · averageSlow=200

●

STAGE 1 · TEMPLATE P1_T021
Selected
fast=2..10/1 · MaxStop=50..400/25 · MaxTPRatio=1.0..4.0/0.25

●

STAGE 1 · WINNING ROW #1024
Selected
fast=5 · MaxStop=125 · MaxTPRatio=2.5

PF	NET	DD	TRADES
2.31	\$18,420	-\$1,840	412

●

STAGE 2 · CHILD TEMPLATE P2_T003 · from P1_T021#1024
Selected
fast=4..6/1 · MaxStop=110..140/5 · MaxTPRatio=2.25..2.75/0.05

●

STAGE 2 · ROW #18
Rejected
fast=4 · MaxStop=110 · MaxTPRatio=2.35

PF	NET	DD	TRADES
2.18	\$16,820	-\$1,740	401

●

STAGE 2 · WINNING ROW #42
Selected
fast=5 · MaxStop=120 · MaxTPRatio=2.55

PF	NET	DD	TRADES
2.48	\$19,240	-\$1,690	418

●

STAGE 3 · RISK TEMPLATE P3_T002 · from P2_T003#42
Selected
ProfitStop=2K..15K/500 · LossStop=2K..10K/500 · MaxTrades=4..8/1

●

STAGE 3 · BEST ROW #87
Current
ProfitStop=8000 · LossStop=4500 · MaxTrades=6

PF	NET	DD	TRADES
2.42	\$18,960	-\$1,480	388

●

STAGE 4 · VALIDATION P4_V004 · from P3_T002#87
Current
Fixed backtest · contract NQ06-26 · walk-forward window

PF	NET	DD	TRADES
2.36	\$18,420	-\$1,520	376

SELECTED NODE

Selected

Stage 2 · row #42

PROFIT FACTOR

2.48

NET P/L

\$19,240

DRAWDOWN

-\$1,690

TRADES

418

PARAMETERS

StartTimeH=8 · Reverse=false

averageFast=5 · averageSlow=200

MaxStop=120 · MaxTPRatio=2.55

Override this branch

↺ Rerun this template

↺ Rerun this stage from scratch

↺ Clone this branch with wider range

⊕ Manually promote a sibling row

⊖ Reject this candidate

FILES

rec_8c2a44 / stage_2 /

└ P2_T003.xml template

└ P2_T003_results.csv 120 rows

└ P2_T003_selected.json row #42

RECIPE COMPLETE · FINAL REVIEW

Deployment-ready candidates

The system completed all 4 stages. 5 candidates passed final validation, 2 with warnings, 1 failed. Approve to package for NinjaTrader deployment.

Export report (PDF)

Compare top 4

Build deployment package

✓ Recipe finished in 3h 18m. 124 templates · 15,420 backtests · 18 winners in Stage 1 → 7 in Stage 2 → 5 in Stage 3 → 4 valid + 1 warning at final validation. Goal of ≥ 4 final candidates reached.

RECOMMENDED

2

valid · no warnings

ACCEPTABLE

2

valid · some warnings

FAILED VALIDATION

1

excluded from package

TOTAL RUNTIME

3h 18m

started 10:51 · finished 14:09

FINAL CANDIDATES

Ranked by DD-adjusted score

All 5Valid 2Warning 2Failed 1

1	Pantheon NQ · Morning Trend #C-04Valid · lineage rec_8c2a44 → S1.b_S08 → ... → S4 Start=08 · Rev=false · Fast=5 · Slow=200 · MaxStop=120 · MaxTPRatio=2.55 · ProfitStop=\$8,000 · LossStop=\$4,500 · MaxTrades=6	PF2.36 NET\$18,420 DD-\$1,520 % DAYS78.4% Inspect lineage Add to package
2	Pantheon NQ · Late Morning #C-03Valid · lineage rec_8c2a44 → S1.b_S08 → ... → S4 Start=08 · Rev=false · Fast=5 · Slow=250 · MaxStop=125 · MaxTPRatio=2.25 · ProfitStop=\$7,500 · LossStop=\$4,000 · MaxTrades=5	PF2.18 NET\$17,110 DD-\$1,620 % DAYS76.1% Inspect lineage Add to package
3	Pantheon NQ · Midday #C-01Warning · lineage rec_8c2a44 → S1.b_S12 → ... → S4 Start=12 · Rev=false · Fast=5 · Slow=175 · MaxStop=150 · MaxTPRatio=2.50 · ProfitStop=\$6,000 · LossStop=\$5,000 · MaxTrades=4 ⚠ % days traded below 70% ⚠ Recent 30-day win % drifted to 54.2	PF2.05 NET\$16,240 DD-\$1,980 % DAYS64.2% Inspect lineage Add to package
4	Pantheon NQ · Early Window #C-02Warning · lineage rec_8c2a44 → S1.b_S04 → ... → S4 Start=04 · Rev=false · Fast=4 · Slow=200 · MaxStop=130 · MaxTPRatio=2.10 · ProfitStop=\$5,500 · LossStop=\$4,000 · MaxTrades=5 ⚠ Settings-contract mismatch · NQ 09-26	PF1.84 NET\$9,120 DD-\$1,620 % DAYS71.4% Inspect lineage Add to package
5	Pantheon NQ · Evening Reversal #C-09Failed · lineage rec_8c2a44 → S1.b_S20 → ... → S4 Start=20 · Rev=true · Fast=6 · Slow=400 · MaxStop=200 · MaxTPRatio=2.00 ⚠ Failed: drawdown \$2,480 > \$2,500 cap ⚠ Boundary parent (Slow=400 was at axis edge)	PF1.42 NET\$6,240 DD-\$2,480 % DAYS48.1% Inspect lineage

DEPLOYMENT PACKAGE

3 of 5 selected · ready to build

What gets handed off to NinjaTrader

FILES

└ C-04_PantheonNQ_Morning.xml
└ C-03_PantheonNQ_LateMorning.xml
└ C-01_PantheonNQ_Midday.xml
└ deployment_report.pdf

PRE-FLIGHT CHECKS

✓ All settings-contract aligned (NQ 06-26)
✓ Position size = 1 on all candidates
✓ Strategy file path verified
⚠ Re-validate before live deployment

SESSION

All artifacts saved under
sessions/rec_8c2a44/
Recipes are immutable once built — re-running creates a new session.